

Maxime Lafleur

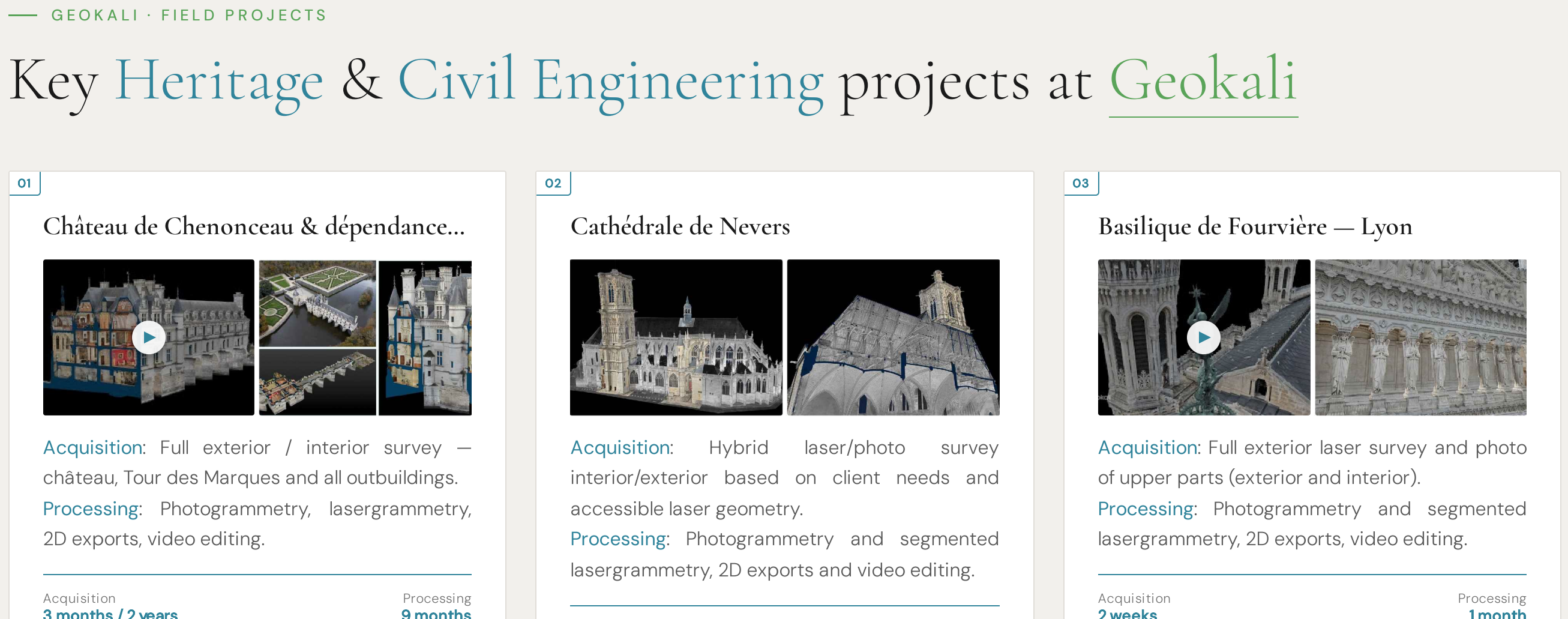
Engineer · Photogrammetrist · Lasergrammetrist · Developer

Passionate about cultural heritage and cutting-edge 3D reconstruction technologies, I initially built my expertise around drone geomatics – from hardware and software development to applied research – gaining full autonomy over the entire process, from study and prototyping to product development (2014–2022).

More recently (2022–2026), my work in the cultural heritage sector immersed me in high-fidelity 3D capture and reconstruction. The technical challenges and the blending of diverse technologies meant mastering specialized skills and methodologies, pushing existing solutions to their limits, and developing automation to eliminate repetitive tasks.

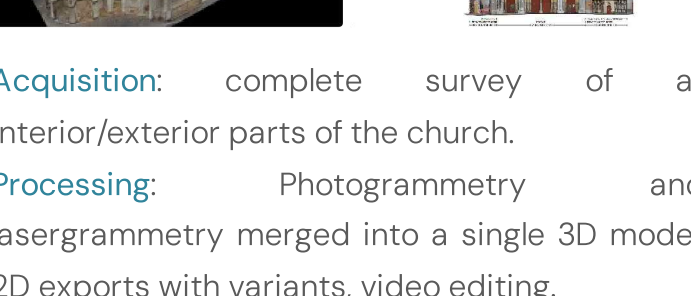
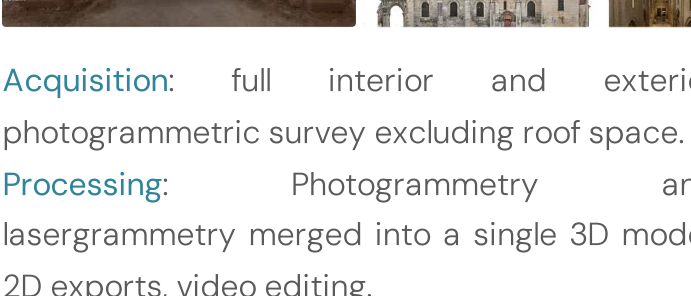
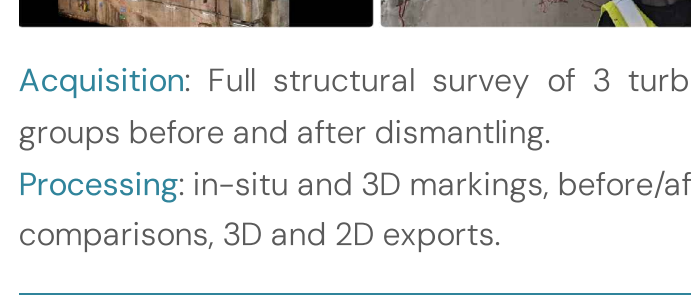
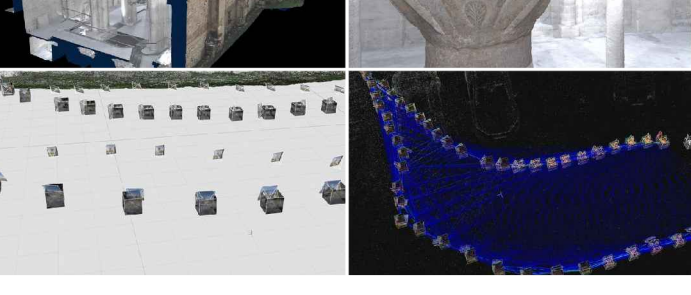
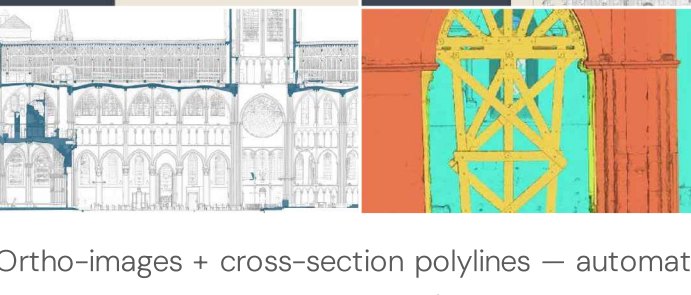
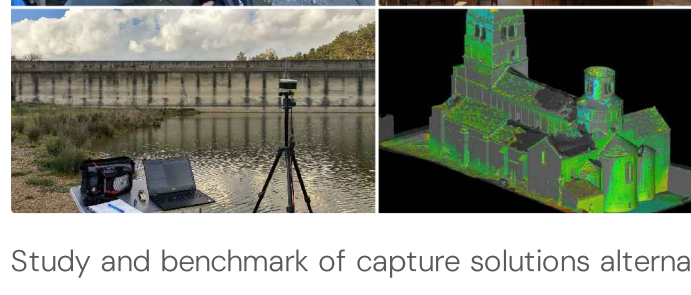
With 3D technology evolving rapidly and opening up new possibilities, I'm eager to keep exploring this field, learn from others and draw on new approaches, as part of a creative team – bringing my 12 years of curiosity and varied experience.

Looking forward to connecting and discussing projects, opportunities and experiences!



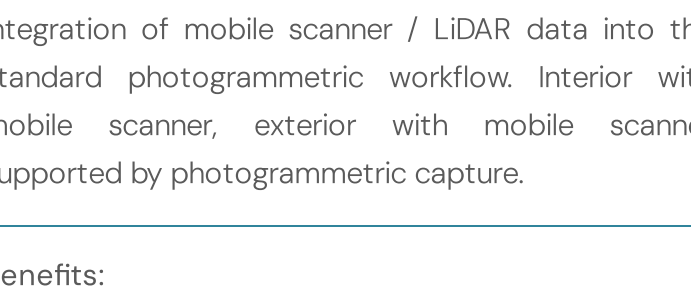
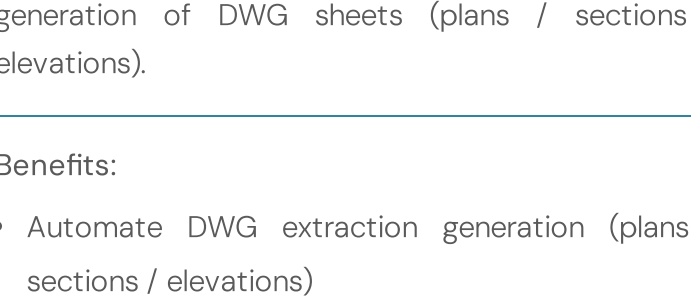
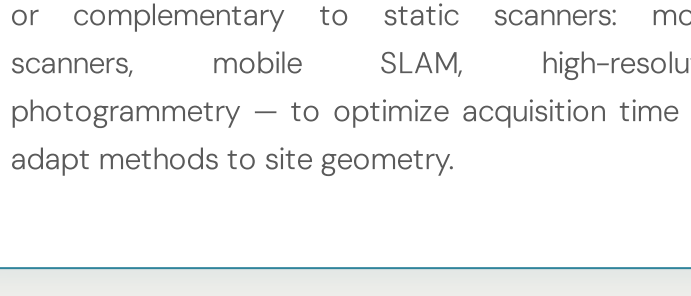
— GEOKALI · FIELD PROJECTS

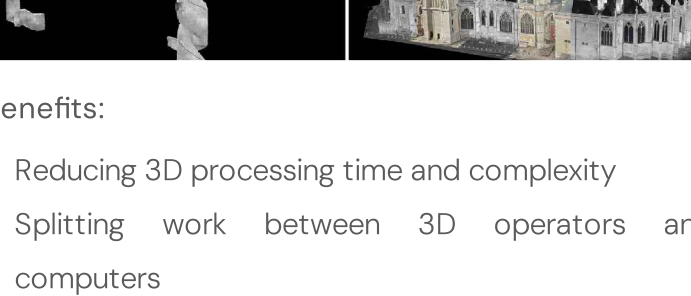
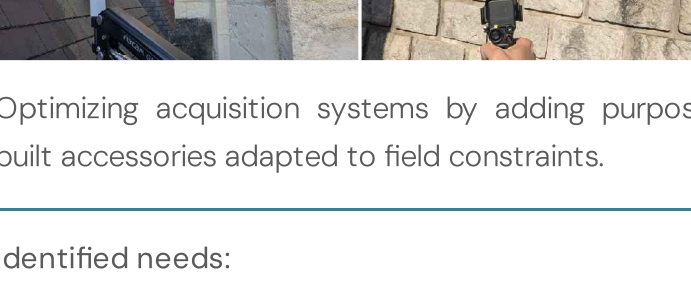
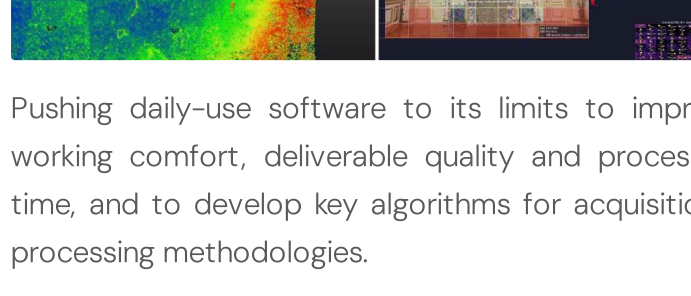
Key Heritage & Civil Engineering projects at Geokali

01 Château de Chenonceau & dépendance..  Acquisition: Full exterior / interior survey — chateau, Tour des Marques and all outbuildings. Processing: Photogrammetry, lasergrammetry, 2D exports, video editing. Acquisition 3 months / 2 years Images ~265 000 3D Models ~80 809M Δ Project size 16 TB Processing 9 months Scans ~5 500 Texture resolution 1mm 2D AutoCAD drawings 173 .dwg ~700 ortho	02 Cathédrale de Nevers  Acquisition: Hybrid laser/photo survey interior/exterior based on client needs and accessible laser geometry. Processing: Photogrammetry and segmented lasergrammetry, 2D exports and video editing. Acquisition 1 month Images ~46 000 3D Models ~30 237M Δ 2D AutoCAD drawings 57 .dwg ~200 ortho Processing 2 months Scans ~1830 Texture resolution 1mm	03 Basilique de Fourvière — Lyon  Acquisition: Full exterior laser survey and photo of upper parts (exterior and interior). Processing: Photogrammetry and segmented lasergrammetry, 2D exports, video editing. Acquisition 2 weeks Images ~38 000 3D Models ~30 399M Δ 2D AutoCAD drawings 57 .dwg ~200 ortho Processing 1 month Scans ~400 Texture resolution 1mm
04 Eglise Notre-Dame — Dijon  Acquisition: complete survey of all interior/exterior parts of the church. Processing: Photogrammetry and lasergrammetry merged into a single 3D model, 2D exports with variants, video editing. Acquisition 3 weeks Images ~54 000 3D Models ~2 30M Δ 2D AutoCAD drawings 40 .dwg ~80 ortho Processing 2 months Scans ~1100 Texture resolution 1mm	05 Eglise Saint-Philibert — Dijon  Acquisition: full interior and exterior photogrammetric survey excluding roof space. Processing: Photogrammetry and lasergrammetry merged into a single 3D model, 2D exports, video editing. Acquisition 2 weeks Images ~30 300 3D Models ~41 38M Δ 2D AutoCAD drawings 4 .dwg ~40 ortho Processing 1 month Scans ~250 Texture resolution 1mm	06 Civil Engineering Structure  Acquisition: Full structural survey of 3 turbine groups before and after dismantling. Processing: in-situ and 3D markings, before/after comparisons, 3D and 2D exports. Acquisition 2x 2 weeks Images ~80 000 3D Models ~3 50M Δ Project size 8 TB 2D AutoCAD drawings 30 .dwg ~100 ortho Processing 9 months Scans ~400 Texture resolution 0.35 mm Defects (Cyclone 3DR) ~900 m

— INNOVATION & DEVELOPMENT

Key Heritage R&D projects at Geokali

07 Mobile LiDAR data integrated into the photogrammetric pipeline  Integration of mobile scanner / LiDAR data into the standard photogrammetric workflow. Interior with mobile scanner, exterior with mobile scanner supported by photogrammetric capture. Benefits: • Maintain a standard 3D production workflow only, replacing static scans with virtual positions from SLAM • Panoramic photo contribution from the SLAM scanner, equivalent to static panoramas Costs: • Georeferencing precision, drift analysis, strict GCP setup — strong mastery of SLAM solutions and their limits • Risk of geometric fidelity failure (drift, double surfaces) with no correction capacity during processing Study site Abbaye de Cluny Ground photo acquisition 2 hours Triangulation SLAM acquisition 2x 45 min Along trajectory 1 meter / 2 m Texture resolution (i ground photogrammetric resolution)	08 Automated 2D DWG sheet generation  Ortho-images + cross-section polylines — automated generation of DWG sheets (plans / sections / elevations). Benefits: • Automate DWG extraction generation (plans / sections / elevations) • Provide an internal operator with an export selection tool for resolutions and client-subscribed options (depth views, contours, confidence map, pure 3D view) • Eliminate any human error in the processing chain • Avoid chained operations during extractions • Systematize deliverables and lock nomenclature, extraction themes and renders • Standardized quick to client requests — extraction productivity multiplied by x30 per dwg sheet Costs: • SDK software development, trial and error with trusted partners, maintenance across new software versions	09 Technology watch  Study and benchmark of capture solutions alternative or complementary to static scans: mobile scanners, mobile SLAM, high-resolution photogrammetry — to optimize acquisition time and adapt methods to site geometry.
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04 3D Production by segmentation  Benefits: • Reducing 3D processing time and complexity • Splitting work between 3D operators and computers • Ability to reprocess a specific area following potential changes • Adapting detail level and texture quality per zone • Offering clients optimized acquisition costs Costs: • Careful cleaning of all laser scans, filtering of images / laser stations, 3D model cleaning, strict file naming conventions.	05 Hardware prototyping  Optimizing acquisition systems by adding purpose-built accessories adapted to field constraints. Identified needs: • Automating confined spaces • Securing the system for specific use cases • Simplifying daily operations (heavy handheld equipment) Adopted solutions: • Laser scanner trolley (productivity gain) • 3D camera gimbal for repetitive facades or confined spaces (DJ Ronin) • Steadycam vest and belt for heavy equipment (Flycam Galaxy) Custom solutions: • Low-profile tripod for terrestrial scanner (daily use) • Camera add-on fixed to terrestrial laser scanner • Custom attachment systems for gimbals	06 Software prototyping  Pushing daily-use software to its limits to improve working comfort, deliverable quality and processing time, and to develop key algorithms for alignment / processing methodologies. Automated script development for: • AutoCAD / Photoshop / Leica Cyclone 3DR / Agisoft Metashape / RealityScan Developed tools: • Multi-scale analysis tool for alignment discontinuities from RealityScan photogrammetric processing • Automated Image QC tool (motion/focal blur, exposure) with renaming, copying and raw data backup • Ortho-image generation with various renders: depth, contours, Xray, 3D confidence, geometric voids — to streamline the drafter's workflow Small algorithms: • Image filtering by location and orientation • WGSR4 to local projected coordinates conversion • Storage redundancy detection script • Automated conversion of panoramic to equirectangular images • Injection of pre-calibrated camera calibrations into EXIF metadata • And other small useful developments on a project-by-project basis
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— PROFESSIONAL EXPERIENCE

Cyclope3D SEPT. 2026 — PRESENT · 1 MO	Geokali 2022 — JUNE 2026 · 3 YRS 9 MO	Microdrones™ / GeoCue 2016 — 2022 · 6 YRS 8 MO	CNRS 2021 — 2022 · 1 YRS 6 MO	Delair 2016 — 2018 · 2 YRS	InventUAV 2015 — 2016 · 1 YRS	Vision du Ciel Int'l 2014 — 2015 · 1 YRS	Groupes SII · Nexter 2013 — 2014 · 1 YRS
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Cyclope3D Software Publisher Cyclope3D · Freelance · Auvergne-Rhône-Alpes, France	1 mo Sept. 2026 — present
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Design and development of **Cyclope3D**, a tool for extracting 2D drawings from 3D data (meshes, point clouds, 3DGS), built for drafters, architects, and laser scanning and photogrammetry surveyors. The tool draws on several years of client feedback and analysis of the deliverables on the market.

- Opening and analysis of 3D deliverables, to get to grips with the survey and check it against the specifications
- Automatic 2D drawing sheet generation, with no manual handling: georeferenced orthomosaics, DXF/DWG drawings and PDF sheets
- Several representations per view (texture, geometry, depth, solid areas, X-ray), overlaid in the drawing
- Guided naming of folders, files and layers, derived from the project and view names
- Customizable deliverables in the company's colors: orthomosaic background, logo

→ [cyclope3d.fr](#)

SOFTWARE DEVELOPMENT	2D DRAWING EXTRACTION	DXF / DWG	ORTHOIMAGES	POINT CLOUDS	3DGS	CULTURAL HERITAGE
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Photogrammetry / Lasergrammetry Engineer Geokali · Full-time · Valence, France	3 yrs Oct. 2022 — June 2026
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3D data acquisition and processing on **cultural heritage and civil engineering** sites, using a full sensor fleet: UAV drones, terrestrial laser scanners (TLS), mobile laser scanners (MLS), DSC cameras and structured light lasers.

→ [See Heritage & Civil Engineering projects](#) · [See R&D projects](#)

Jean-Marc Anneron — Drafterman-Surveyor-Laser scanning specialist, historic monuments since 2001

“I had the pleasure of working alongside Maxime Lafleur for several years on demanding operations related to surveying and graphical restitution of historic monuments. Looking back (and as a Heritage drafterman with over 25 years of experience), I can say without hesitation that he is among the best specialists in the field! I have had the chance to collaborate with Maxime on an extremely rare profile, combining technical ingenuity, adaptability, creativity and rigor. Where many see a blocker or a constraint, he already starts designing a solution: he codes, develops routines, automates complex tasks and creates efficient methodologies that push 3D/2D tool outputs to their maximum. The quality, readability and ease of use of his deliverables reflect this. In the field, his mastery of capture operations is remarkable. Whether in 3D digitization (photogrammetry laser scanning), he shows great rigor and an excellent reading of situations, which perfectly matches and feeds Geokali's expertise. But beyond the skills, what I probably appreciate most about him is his human quality: he brings to a collaboration. Maxime is approachable, reliable, attentive, proactive and generously involved. When working with him, you immediately feel you can move forward with confidence, even on the most complex projects.”

PHOTOGRAMMETRY	LIDAR TLS/MLS	UAV	3D RECONSTRUCTION	LEICA SDK	DWG AUTOMATION	POINT CLOUDS	HARDWARE PROTOTYPING
CULTURAL HERITAGE	R&D						

UAV LiDAR Engineer Microdrones™ / GeoCue · R&D · Toulouse	6 yrs May 2016 — Sept. 2022
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Within the R&D department of Microdrones / GeoCue, responsible for developing **new UAV-embedded LiDAR prototypes**, designing high-precision LiDAR targets, and QC of new processing software.

- Operational setup of Microdrones France subsidiary (office design, construction follow-up, HR)
- Hardware design of UAV LiDAR systems
- Field testing and data QC
- Hardware and software benchmarks
- Development of an automated tree counting tool
- Automated pointline classification
- Parachute system development and EASA certification approval

Vincent Eucat — Head of Operations “I had the opportunity to work alongside Maxime Lafleur during my time at Microdrones, and I can wholeheartedly vouch for his excellent skills as an engineer. As part of our drone data acquisition missions for various industrial clients in France, Maxime was a true technical pillar. His analytical mindset and great resourcefulness allowed him to efficiently diagnose and solve complex hardware and software issues, often under demanding field conditions. Whether it was last-minute adjustments to the equipment or optimizing flight operations, his ingenuity always ensured the continuity and quality of our operations. I highly recommend him to any organization looking for a reliable and talented field engineer.”	Erwan Gavaille — Founder of geopilot “I had the pleasure of working with Maxime on the development of drone payloads. He was in charge of the electronics and project management while I worked on the mechanics. It was my first job in this field, but he trusted me, taught me and challenged me throughout the projects, particularly appreciated his skills, his involvement and his friendliness in driving the projects to success. He also managed a large part of the organization and setup of our offices in France. I highly recommend Maxime for his professionalism, his good mood and his ability to support the people he works with.”
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UAV LIDAR R&D	HARDWARE DESIGN	EASA CERTIFICATION	POWERLINE CLASSIFICATION	TREE COUNTING	FIELD TESTING
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3D / GIS Support for Antarctic PhD Project CNRS · Civil Service · Remote	1.5 yrs Feb. 2021 — Jul. 2022
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Technical and methodological support for a **PhD project** on the **spatial distribution of penguins in Antarctica**. 3D processing of 11 Antarctic sites.

- GNSS base station PPP post-processing
- Multi-LiDAR processing
- PPP-like photogrammetry
- High-precision Ortho-DEM generation

ANTARCTICA	PPK LIDAR	GNSS PPP	PHOTOGRAMMETRY	ORTHO-DEM	GIS
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UAV Engineer Payload Integration & Control/Command Delair · Full-time · Toulouse	2 yrs May 2016 — May 2018
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Within Delair-Tech's R&D department, under Microdrones' strategic direction, responsible for:

- Control/command system development for Microdrones UAVs
- Integration of Delair-Tech payloads (multispectral, optronic, photogrammetry) on Microdrones m4-1000
- Viaculture research project in partnership with Naio Technologies (vineyard row mapping with PPK)
- Other prototyping projects

Christophe Eucat — Head of Operations

“I worked with Maxime at Delair on several tests and projects related to drones and imaging. Friendly and particularly rigorous, he enjoys working as part of a team. We also share a particular interest in several areas outside of work, which only strengthened our collaboration.”

Pierre-Louis Tilak — Director at Titakio

“I had the chance to work for several years with Maxime Lafleur when he was in charge of integrating Microdrones products into the Delair ground station, as well as developing and integrating several payloads, including multi-spectral sensors. He is capable of managing application software development, embedded development, as well as handling flight tests, operations, and processing of photogrammetry data, showing a thorough rigor and a complete skill set. He is an extremely motivated, reliable and passionate person who, once he takes part in a project, won't let go until it works. I strongly recommend him, as with over 10 years of experience he will be a solid ally for your projects.”

PAYLOAD INTEGRATION	FIXED-WING UAV	EMBEDDED SYSTEMS
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Co-Founder InventUAV · Open-Source Startup · France	1 yr May 2015 — May 2016
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Co-founder of an open-source UAV startup developing an **autonomous indoor drone** for warehouse management and inventory.

- ArduPlane plugin for Gazebo / ROS
- ArduCopter debugging
- Visual recognition for indoor missions

ARDUPILOT	ROS / GAZEBO	OPEN SOURCE	INDOOR DRONE
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UAV Apps Engineer (GCS / GUI) Vision du Ciel Int'l · Orange	1 yr Jul. 2014 — Jun. 2015
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Development of **UAV mission control** software solutions on Windows: mission planning, control, supervision and reporting.

- GUI development in C#, JS, Python
- Embedded development in C++
- Open-source and industrial protocol integration
- Prototyping: UAVs, radio controllers, gimbals, parachutes

GCS DEVELOPMENT	C# / PYTHON / JS	PIXHAWK	HARDWARE PROTOTYPING
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System Architect Nexter Systems Groupes SII · Satory	1 yr Oct. 2013 — Jul. 2014
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Writing, analysis and capitalisation of system requirements for Nexter. Architecture in IBM DOORS, document traceability, automation script development, object-oriented C++ modelling.

SYSTEM ARCHITECTURE	IBM DOORS	C++	REQUIREMENTS ANALYSIS
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More information available on request

— TECHNICAL EXPERTISE

Skills & tools (non exhaustive)

O1 2020	Patent : 3D Accuracy Star - Catenary Lines	PATENT
O2 2020	Validation of LiDAR Survey Data by Comparison of Several Uncertainty Models	PUBLICATION
O3 2020	Comparison of UAV LiDAR Odometry of Rotating and Fixed Velodyne Platforms	PUBLICATION
O4 2019	Denoising of 3D Point Clouds	PUBLICATION
O5 2012	Tri-Rotors UAV Stabilization for Vertical Takeoff and Hovering	PUBLICATION

EDUCATION

Education & International exchanges

EU	 ECE Paris Master of Science (MSc) Electrical & Electronics Engineering 2008–2013	US	University of California, Irvine Exchange Program Global Management & Manufacturing 2012	CA	Concordia University Semester Exchange Engineering & Computer Science 2011
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CONTACT

Let's connect !

— RESEARCH & INNOVATION

Publications & Patents

01 2020 Patent : 3D Accuracy Star + Catenary Lines	PATENT
02 2020 Validation of LiDAR Survey Data by Comparison of Several Uncertainty Models	PUBLICATION
03 2020 Comparison of UAV LiDAR Odometry of Rotating and Fixed Velodyne Platforms	PUBLICATION
04 2018 Denoising of 3D Point Clouds	PUBLICATION
05 2015 Tri-Rotors UAV Stabilization for Vertical Takeoff and Hovering	PUBLICATION

— EDUCATION

Education & International exchanges

EU European Drone Certificate Open A (B2/AD and specific CATS) 2026	ECE Paris Master of Engineering (MEng) Electrical & Electronics Engineering 2008–2013	US University of California, Irvine Exchange Program Global Management & Manufacturing 2012	CA Concordia University Semester Exchange Engineering & Computer Science 2011
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— CONTACT

Let's connect !

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LANGUAGES French (native) · English (Full Professional)
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